

SARMELLEK, Hungary

WMO: 129220

Lat: 46.683N

Lon: 17.167E

Elev: 125

StdP: 99.83

Time Zone: 1.00 (EUC)

Period: 01-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-10.1	-7.7	-14.0	1.1	-8.9	-11.2	1.5	-6.2	11.9	1.3	10.8	-0.2	3.3	0	0.498

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.8	32.6	21.9	31.0	21.3	29.4	20.6	23.1	30.1	22.2	28.9	21.3	27.9	3.2	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
20.9	15.8	26.9	20.0	14.9	25.6	19.1	14.1	24.6	69.0	29.9	65.7	28.9	62.5	28.0	26.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.6	8.9	7.5	-13.7	35.3	3.9	2.0	-16.5	36.8	-18.8	38.0	-21.0	39.1	-23.9	40.6		
(5)			-14.4	24.8	3.8	0.9	-17.2	25.5	-19.4	26.0	-21.5	26.5	-24.3	27.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	11.3	0.1	2.1	6.4	11.9	15.7	19.9	21.6	21.4	16.2	11.4	6.2	1.6		
	DBStd	8.40	4.40	4.29	4.45	3.55	3.51	3.43	3.10	2.95	3.25	3.82	3.99	4.02		
	HDD10.0	1092	307	222	124	21	2	0	0	0	1	31	123	261		
	HDD18.3	2894	565	455	369	193	95	23	8	8	81	217	363	517		
	CDD10.0	1552	0	1	13	79	179	297	359	352	186	73	11	2		
	CDD18.3	311	0	0	0	1	13	69	109	102	16	1	0	0		
	CDH23.3	3233	0	0	0	21	173	716	1122	1020	169	12	0	0		
	CDH26.7	1113	0	0	0	2	29	226	427	391	38	0	0	0		
(14) Wind	WSAvg	2.9	2.9	3.2	3.4	3.4	3.2	2.9	2.8	2.5	2.7	2.5	2.6	2.7		
(15) Precipitation	PrecAvg	667	34	33	42	52	59	75	82	72	56	44	72	46		
	PrecMax	1036	63	85	111	157	152	140	190	175	104	141	176	99		
	PrecMin	417	2	8	13	4	2	7	8	13	10	1	20	7		
	PrecStd	157	14	19	23	38	36	35	54	46	29	39	41	22		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	13.2	15.7	21.0	25.7	29.4	33.2	35.0	35.5	30.3	25.1	19.2	14.9		
		MCWB	9.0	9.7	12.2	16.8	18.8	21.9	22.0	21.5	20.3	17.7	14.1	11.3		
	2%	DB	10.5	12.8	18.8	23.3	27.2	31.3	32.5	32.8	27.5	22.3	16.2	11.9		
		MCWB	7.3	8.2	11.2	14.4	18.6	21.6	22.2	21.7	19.1	16.7	12.6	9.3		
	5%	DB	8.2	10.7	16.4	21.6	25.3	29.4	30.9	30.7	25.2	20.4	13.9	9.1		
		MCWB	5.9	7.1	10.1	13.7	17.8	20.7	21.3	21.1	18.1	15.5	11.1	6.8		
	10%	DB	6.1	8.3	13.9	19.5	23.3	27.6	29.1	28.8	23.0	18.0	12.2	6.9		
		MCWB	4.3	5.6	8.8	12.7	16.6	19.8	20.7	20.5	17.0	14.2	10.2	5.2		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	9.6	10.3	12.8	17.3	20.6	23.9	24.6	24.0	21.1	18.4	14.5	11.5		
		MCDB	12.8	15.4	19.1	24.7	27.1	30.7	31.2	30.8	28.4	23.7	18.5	14.5		
	2%	WB	7.7	8.6	11.6	15.5	19.4	22.3	23.2	22.9	19.8	17.0	12.9	9.4		
		MCDB	10.0	12.3	17.5	22.0	25.7	28.8	30.1	30.1	25.9	21.7	15.7	11.8		
	5%	WB	6.1	7.3	10.4	14.2	18.2	21.3	22.2	22.1	18.7	15.8	11.6	7.2		
		MCDB	7.9	10.0	15.6	19.8	24.1	28.0	29.0	28.8	24.1	19.7	13.7	8.8		
	10%	WB	4.5	6.0	9.3	13.2	17.1	20.4	21.3	21.1	17.5	14.7	10.2	5.5		
		MCDB	5.9	7.9	13.4	18.8	22.4	26.6	27.8	27.5	21.9	17.6	11.9	6.6		
(35) Mean Daily Temperature Range	5% DB	MDBR	6.8	7.6	10.0	11.4	11.3	11.5	11.8	12.1	10.8	10.2	7.5	6.2		
		MCDBR	10.6	12.1	14.6	15.4	14.7	14.3	14.7	15.6	14.6	13.3	10.9	9.6		
	5% WB	MCWBR	7.7	7.9	8.3	8.1	7.2	6.1	5.8	6.2	6.9	8.0	7.5	7.1		
		MCWBR	10.1	11.0	13.3	13.7	13.3	13.3	13.5	13.8	12.9	12.7	10.3	9.4		
(40) Clear-Sky Solar Irradiance	taub		0.329	0.354	0.389	0.437	0.428	0.431	0.436	0.440	0.418	0.388	0.357	0.329		
	taud		2.379	2.324	2.251	2.133	2.212	2.244	2.230	2.230	2.259	2.323	2.362	2.394		
	Ebn at Noon		752	805	827	818	840	838	827	807	787	753	706	706		
	Edh at Noon		73	94	117	144	138	134	135	129	115	93	74	65		
(44) All-Sky Solar Radiation	RadAvg		1.18	1.97	3.17	4.52	5.51	6.01	5.95	5.27	3.71	2.44	1.30	0.89		
	RadStd		0.15	0.29	0.38	0.45	0.48	0.43	0.42	0.54	0.45	0.26	0.15	0.13		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (71 neighbors)	+0.55	N/A	N/A	N/A	+0.48	+0.45	-110	-152	+92	+54

Nomenclature: See separate page